

LIST

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LIST

A list is a versatile and mutable data structure in Python that is used to store multiple items in a single variable. Items in a list are ordered, changeable, and allow duplicate values.

Syntax:

```
list_name = [item1, item2, item3, ...]
```

Example:

```
list1=["apple", "banana", "cat"]
```

LIST

Key Rules:

1. Square Brackets: Lists are enclosed in `[ ]`.
2. Comma-Separated: Items in the list are separated by commas.
3. Can Be Empty: You can create an empty list using `[]`.
4. Elements Can Be Any Data Type: A list can include integers, strings, floats, and even other lists.

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Key Characteristics of Lists:

- 1.Ordered: The order of elements in a list is preserved.
- 2.Mutable: Elements in a list can be changed after the list is created.
- 3.Allows Duplicates: Lists can have multiple items with the same value.
- 4.Heterogeneous: Lists can store elements of different data types.

LIST EXAMPLE

Empty list

```
empty_list = []
```

List of integers

```
numbers = [1, 2, 3, 4, 5]
```

List of strings

```
fruits = ["apple", "banana", "cherry"]
```


LIST EXAMPLE

Mixed data types

```
mixed = [42, "hello", 3.14, True]
```

Nested list

```
nested = [[1, 2], [3, 4]]
```

ITERATING THROUGH A LIST

Using for Loop:

```
for fruit in fruits:  
    print(fruit)
```

Using Indexes:

```
for i in range(len(fruits)):  
    print(fruits[i])
```

ACCESSING LIST ELEMENT

A. Indexing:

```
fruits = ["apple", "banana", "cherry"]  
print(fruits[0])
```

B. Slicing:

```
numbers = [1, 2, 3, 4, 5]  
print(numbers[1:4])
```


LIST OPERATION

a. Adding Elements:

```
fruits = ["apple", "banana"]  
fruits.append("cherry")      # Adds at the end  
fruits.insert(1, "blueberry") # Adds at index 1  
print(fruits) # Output: ['apple', 'blueberry', 'banana', 'cherry']
```

LIST OPERATION

b. Removing Elements

`fruits.remove("banana")` # Removes first occurrence of "banana"

`fruits.pop(1)` # Removes item at index 1

`fruits.clear()` # Removes all items

c. Modifying Elements

`fruits[0] = "kiwi"`

`print(fruits)` # Output: ['kiwi', 'banana']

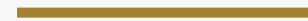
QUESTIONS

- a. Write a Python program to count the number of even and odd numbers in a given list.
- b. Write a Python function that takes a list of numbers and returns the sum of all the elements.
- c. Write a function that takes a list of integers and returns the maximum value in the list.
- d. Write a Python program that takes a list of numbers and returns the square of all elements.

Input: [2, 4, 3, 7]

Output: [4, 16, 9, 49]

THANK YOU



FOR COMING